

Advanced Self-Supervised Learning and JEPA Models

Naeemullah Khan

naeemullah.khan@kaust.edu.sa



جامعة الملك عبد الله
للعلوم والتقنية

King Abdullah University of
Science and Technology

KAUST Academy
King Abdullah University of Science and Technology

October 6, 2026

1. Recap: from contrastive and self-distillation SSL to latent prediction
2. Evolution of SSL and a three-way taxonomy (invariant, generative, JEPA)
3. Background: masked autoencoders, the collapse problem, and why SSL transfers
4. JEPA motivation: world models and objective-driven AI
5. I-JEPA: architecture, masking, objective, training, and efficiency
6. Evaluating SSL encoders: probes, low-shot, and fine-tuning protocols
7. From images to video: V-JEPA (masking, probes); **V-JEPA 2** (scale, LLM alignment, planning)
8. Specialized extensions: Point-, CNN-, VL-, MC-, A-JEPA; video efficiency (SALT)
9. Limitations, open problems, and further study

- ▶ **Pretext tasks** (rotation, jigsaw, colorization): free supervision from the data itself, but features are only as good as the proxy task.
- ▶ **Contrastive learning** (instance discrimination, SimCLR, MoCo): pull augmented views together and push *negatives* apart; requires large batches or memory queues.
- ▶ **Self-distillation without negatives** (BYOL, DINO, DINOv2): a **student** matches a **teacher** that is an **exponential moving average (EMA)** of the student, with no negative pairs at all.
- ▶ **iBOT**: adds **masked image modeling** to self-distillation; the student predicts the teacher's *tokens* for hidden patches.

Today: predict abstract representations of unseen or future content, and use the resulting predictor as a **world model**.

- ▶ **iBOT's token-level distillation** is already *latent* masked prediction: hide patches, then predict the **teacher's representation** of the missing content instead of its pixels.
- ▶ **data2vec** (Baeviski et al., 2022) generalized this across speech, vision, and language: always predict **contextualized latent targets** from an EMA teacher.
- ▶ **JEPA** (this lecture) turns the recipe into an explicit architecture:
 - a **context encoder** sees only the visible input;
 - a separate **predictor** maps context $\rightarrow$ predicted target embeddings;
 - an **EMA target encoder** provides the targets (as in BYOL / DINO / iBOT).
- ▶ New emphasis: the predictor acts as a **world model**, a module that reasons about unseen or future states in representation space.

- ▶ **Early methods:** rotation prediction, jigsaw, colorization. Cheap supervision, limited semantic depth.
- ▶ **Joint-embedding SSL (contrastive and non-contrastive):** SimCLR, MoCo (negatives); BYOL, VICReg (no negatives; VICReg instead regularizes embedding variance and covariance). Align embeddings across views while avoiding collapse.
- ▶ **Masked modeling / generative SSL:** MAE, BEiT. Reconstruct tokens or pixels in observation space.
- ▶ **Latent-target masked prediction:** iBOT (2022), data2vec (2022). Mask the input and predict an EMA teacher's *representations*; the direct precursor to JEPA.
- ▶ **Predictive in latent space (JEPA):** I-JEPA, V-JEPA. Predict *representations* of held-out regions rather than raw observations, with an explicit predictor module.

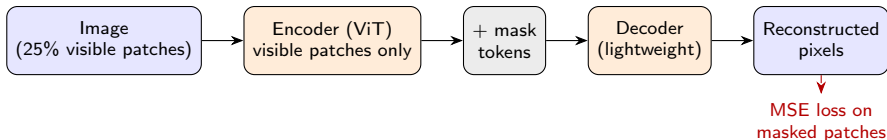
Cross-method survey of SSL for ViTs: [arXiv:2408.17059](https://arxiv.org/abs/2408.17059).

Three Architectural Families in Self-Supervised Learning

Family	Core mechanism	Representative examples
Joint-embedding (<i>invariant</i>)	Pull together compatible views; contrastive or regularized losses prevent collapse.	SimCLR, MoCo, BYOL, VICReg
Generative (<i>reconstruction</i>)	Fill in missing input in pixel or token space; model must explain fine observation detail.	MAE, BEiT, VAEs, GAN-style objectives
JEPA (<i>latent prediction</i>)	Predict the embedding of a target region from context embeddings; captures semantics without a full pixel likelihood.	I-JEPA, V-JEPA, V-JEPA 2, Point-JEPA, CNN-JEPA

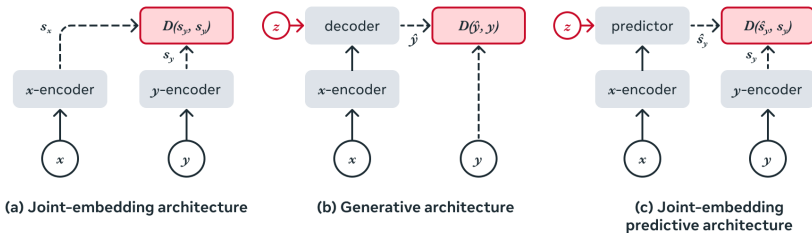
- ▶ Exploits **structure** in natural data (spatial context, temporal continuity, language statistics).
- ▶ Contrastive methods stress **invariances** to augmentations; reconstruction methods force **complete** explanations of the input; JEPA emphasizes **predictive** structure in **representation space**.
- ▶ Strong **linear / low-data probes** suggest encoders expose semantically useful axes for downstream tasks.
- ▶ Active research: theoretical limits, failure modes (shortcuts), and unified views across the three families above.

- ▶ **MAE** (He et al., 2022): mask a large fraction ($\sim 75\%$) of image patches; the encoder sees **only visible patches**; a lightweight decoder reconstructs the **pixels** of the masked ones.
- ▶ Loss: mean squared error **in pixel space**, computed on masked patches only.
- ▶ Asymmetric design (big encoder, small decoder) makes pre-training cheap and scalable.



- ▶ **Limitation:** the model must explain *every* pixel, including unpredictable texture and noise. JEPA treats this as wasted capacity.

- ▶ **(a) Joint-embedding (invariance):** map views x, y to embeddings and pull compatible pairs together; can bias toward invariance shortcuts.
- ▶ **(b) Generative:** decode $\hat{y}$ from context x (and latent z), with the loss in **pixel/input space**; must explain unpredictable detail.
- ▶ **(c) JEPA:** predict the **embedding** $\hat{s}_y$ of y from context and compare to s_y from a target encoder; semantic targets discard irrelevant variability.

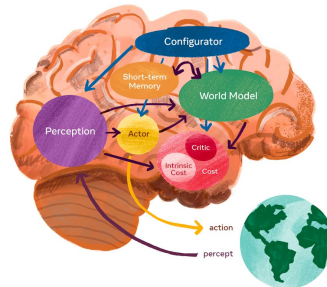


Reference: Adapted from Meta I-JEPA blog.

- ▶ **Representation collapse:** if both encoders output a *constant* vector for every input, any similarity or prediction loss is minimized without learning anything.
- ▶ Three families of **anti-collapse mechanisms** (covered earlier):
 - **Contrastive negatives:** push non-matching pairs apart (SimCLR, MoCo).
 - **Information regularization:** keep per-dimension **variance** up and **covariance** down (VICReg, Barlow Twins).
 - **Architectural asymmetry:** **stop-gradient** on the target branch + **EMA teacher** (BYOL, DINO, iBOT); BYOL adds a student-side **predictor**, DINO **centers and sharpens** teacher outputs.
- ▶ **JEPA inherits the third recipe:** the target encoder is an EMA copy (no gradients flow through targets), and the narrow predictor breaks the symmetry between branches.
- ▶ Contrastive and non-contrastive objectives are closely related in theory (Garrido et al., 2023), but a full account of *why* EMA plus asymmetry suffices remains an open question.

- ▶ Most self-supervised learning (SSL) approaches focus on predicting local features or reconstructing pixels. Can we go further and predict high-level, abstract embeddings instead?
- ▶ **JEPA's key idea:** Predict the *representations* of masked or missing regions, not just raw pixel values. This approach enables semantically rich, meaningful representations.
- ▶ JEPA and world models are proposed as a path to machines that internalize how the world works. By predicting abstract representations rather than details, models can focus on semantics and ignore unpredictable noise.
- ▶ This shift reflects a broader vision (Y. LeCun et al.): powerful architectures like Transformers, while impressive, still lack the human/animal ability to acquire deep understanding from little labeled data. Purely scaling models does not guarantee human-like world understanding.

- ▶ A modular stack for autonomous intelligence (expository diagram):
Configurator routes tasks; **Perception** ingests sensors; **World model** predicts states and fills in missing information; **Actor** proposes actions; **Cost** (intrinsic + critic) scores outcomes; **short-term memory** holds recent interaction history.
- ▶ **SSL** is meant to learn these components from unlabeled streams, analogous to observing the world without a label for every frame.
- ▶ **Abstract representations** focus on what matters and suppress noise (cf. selective attention / inattentional blindness in human perception).



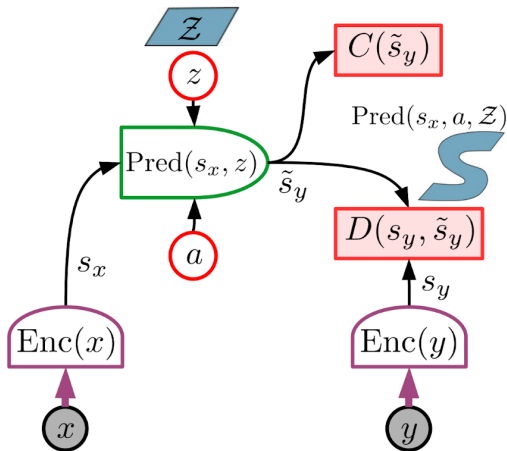
Conceptual AMI diagram (figure as used in Turing Post; image credit Meta AI / Y. LeCun).

What Are Joint Embedding Predictive Architectures (JEPA)?

The model learns by predicting the **embedding** of one part of the input (e.g. a future segment or a masked region) from the representation of another (context / past), using paired encoders and a predictor. For example, predict the embedding of a masked image region (target) from the visible (context) regions.

- ▶ Do not reconstruct pixels; learn on the level of abstract representations.
- ▶ The primary training signal is in **representation space**.
- ▶ Concrete models include I-JEPA (images), V-JEPA (video), MC-JEPA (motion/content), and extensions to other modalities.

- **Notation:** x = observed past/present; y = future (or masked target); a = action; z = latent variable (what x does not determine); $D()$ = prediction cost.
- Encoders map x, y to representations s_x, s_y .
- The **predictor** maps (s_x, a, z) to $\tilde{s}_y$; training minimizes $D(s_y, \tilde{s}_y)$ in representation space.



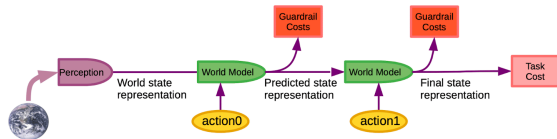
Y. LeCun, "Architecture for the world model: JEPA," slide 37, Ding-Shum Lecture, Harvard University, 2024 (as reproduced in Turing Post).

- ▶ The same world model can be **unrolled** over time: each predicted state feeds the next step, with actions $a_0, a_1, \dots$ and **guardrail** / task costs on the trajectory.
- ▶ Conceptually related to **model predictive control**: optimize actions against predicted costs under learned dynamics in representation space.
- ▶ The same idea appears later in **V-JEPA 2-AC** for robot planning.

Objective-Driven AI: Multistep/Recurrent World Model

Y. LeCun

- ▶ Same world model applied at multiple time steps
- ▶ Guardrail costs applied to entire state trajectory
- ▶ This is identical to Model Predictive Control (MPC)
- ▶ Action inference by minimization of the objectives
 - ▶ Using gradient-based method, graph search, DP, MCTS,....



“Objective-Driven AI: Multistep/Recurrent World Model.” Y. LeCun, slide 42, LMU München, 2023 (as reproduced in Turing Post).

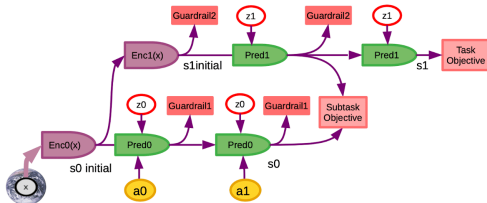
- ▶ Higher levels can predict **longer horizons** in **more abstract** representations; lower levels run on shorter time scales with concrete actions.
- ▶ Higher-level predictions can define **subtask objectives** for lower levels; **guardrails** at each level encode safety or constraints.
- ▶ Hierarchical JEPA remains largely a **proposal** (LeCun, 2022); current systems implement only the single-level version.

Objective-Driven AI: Hierarchical Planning

Y. LeCun

▶ Hierarchical World Model and Planning

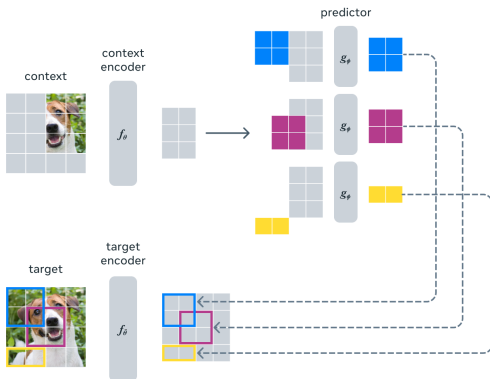
- ▶ Higher levels make longer-term predictions in more abstract representations
- ▶ Predicted states at higher levels define subtask objectives for lower level
- ▶ Guardrail objectives ensure safety at every level



“Objective-Driven AI: Hierarchical Planning.” Y. LeCun, slide 43, LMU München, 2023 (as reproduced in Turing Post).

- ▶ Introduced by Meta AI (Assran et al., 2023) as an instantiation of Yann LeCun's **Joint Embedding Predictive Architecture** vision: learn an *internal model* of the world by comparing **abstract representations** rather than raw pixels.
- ▶ Self-supervised from unlabeled images; targets are **feature embeddings** of masked regions predicted from visible context, not pixel reconstruction.

- ▶ Generative objectives often try to explain **every** missing detail even when the world is **stochastic** at pixel level, wasting capacity on irrelevant variation (Meta blog example: hands in generative models).
- ▶ Predicting **embeddings** steers the model toward **stable, high-level** structure: objects, parts, and layout.
- ▶ Empirically: strong **linear probing** and **semi-supervised** transfer; competitive on some **low-level** tasks (e.g. counting, depth) despite simpler augmentation than many contrastive methods.



Reference: Meta I-JEPA blog architecture.

- ▶ **Context encoder (f_θ):** ViT on **visible** patches only.
- ▶ **Predictor (g_ϕ):** narrow ViT that maps context tokens + target **positional** information to predicted embeddings for each target block.
- ▶ **Target encoder ($f_{\bar{\theta}}$):** encodes full image for targets; weights track the context encoder via **exponential moving average (EMA)**, not gradient through targets.
- ▶ Training minimizes distance between predicted and target **embeddings**.

- ▶ SSL objective lives in **embedding space**: the model predicts **representations** of target blocks, not pixels.
- ▶ **Targets**: sample $M=4$ large blocks per image (each $\sim 15\text{--}20\%$ of the image, varied aspect ratio), big enough to carry **semantic** content, unlike tiny random patches.
- ▶ **Context**: one large informative block ($\sim 85\text{--}100\%$ scale) with any overlapping target regions **removed**, so no target content is visible in the input.
- ▶ The predictor is a **restricted world model** for **static** images: it reasons about what is plausible in unseen regions given partial observation.
- ▶ Mask design is the **core inductive bias** of this family; it replaces the hand-crafted augmentations of contrastive methods.

Loss (representation space only): average L_2 distance between predicted and target patch embeddings over the M target blocks B_i :

$$\mathcal{L} = \frac{1}{M} \sum_{i=1}^M \sum_{j \in B_i} \|\hat{s}_y(j) - s_y(j)\|_2^2, \quad \hat{s}_y(j) = g_\phi(f_\theta(x_{\text{ctx}}), m_i)$$

where g_ϕ is the predictor, f_θ the context encoder, and m_i the target block's **positional** tokens.

- **Targets** $s_y(j) = f_{\bar{\theta}}(y)$ come from the target encoder; **no gradient** flows through them (stop-gradient).
- **EMA update** of the target encoder (as in BYOL, DINO, and iBOT):

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta, \quad \tau \nearrow 1 \text{ over training.}$$

- There are no negatives and no pixel decoder; the asymmetry (predictor + EMA) prevents collapse.

- ▶ Sample multi-block masks; for each target block, predict its embedding from context tokens and target position encodings.
- ▶ **No multi-view contrastive pipeline:** unlike many SSL methods, I-JEPA avoids expensive pipelines that process many augmented views per step. Only context blocks need the context encoder pass, and the target path uses a single view.
- ▶ Feature-space loss + EMA target network for stable targets; avoids collapse without pixel reconstruction.
- ▶ Mask design (large semantic targets, informative context) matters more than hand-crafted view augmentations for this family.

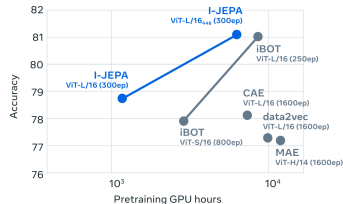
- ▶ To **visualize** what the predictor represents, Meta trains a **stochastic decoder** that maps predicted embeddings toward **pixel sketches** inside a blue box (for analysis only, not the training objective).
- ▶ The model tends to capture **object parts** and **pose** (e.g. dog's head, wolf legs, building structure) rather than arbitrary texture noise.



For each image, everything outside the blue box is encoded and given to the predictor as context. A separately trained generative decoder sketches the contents represented by the predictor's output inside the box. The predictor clearly captures the semantics of what should be filled in (the top of the dog's head, the wolf's legs, the other side of the building). Adapted from the Meta I-JEPA blog.

- ▶ On ImageNet-1K **linear evaluation**, I-JEPA reaches strong accuracy at fewer **pretraining GPU-hours** than several masking and reconstruction baselines (e.g. MAE, data2vec) in Meta's reported comparison.
- ▶ Intuition: representation prediction avoids heavy per-pixel decoder compute and focuses learning on features that transfer.

ImageNet-1K linear evaluation



ImageNet-1K linear eval vs. GPU hours (Meta).

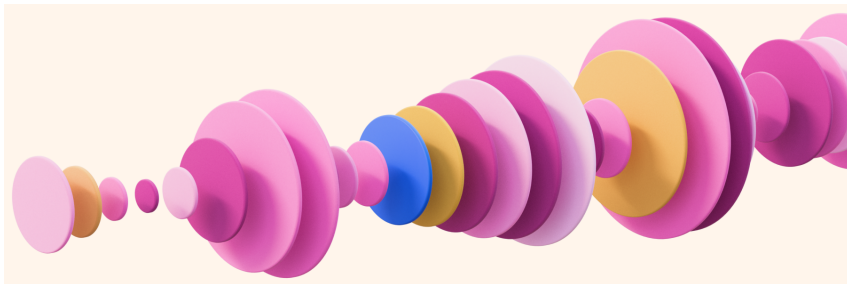
- ▶ **Frozen evaluation:** pre-trained encoder weights are **never updated**; only a small head is trained on top. Measures representation quality directly.
 - **Linear probe:** a single linear layer on pooled features.
 - **Attentive probe:** a small cross-attention layer pools patch/token features before the classifier. Standard for video models (used by V-JEPA); more expressive than a linear probe but still cheap.
- ▶ **Low-shot / semi-supervised:** same idea, but with only e.g. 1%, 5%, or 10% of the labels, which tests **sample efficiency**.
- ▶ **Full fine-tuning:** all weights updated on the downstream task. Strongest numbers, but expensive, and no longer a measure of the *pre-trained* representation alone.
- ▶ JEPA-family papers emphasize **frozen + low-shot** protocols; keep this in mind when comparing to methods tuned for full fine-tuning (e.g. MAE).

Method	Architecture	Epochs	Top-1 accuracy
<i>Methods without view data augmentations</i>			
data2vec	ViT-L/16	1600	73.3
MAE	ViT-L/16	1600	67.1
	ViT-H/14	1600	71.5
I-JEPA	ViT-L/16	600	69.4
	ViT-H/14	300	73.3
	ViT-H/16 ₄₄₈	300	77.3
<i>Methods using extra view data augmentations</i>			
iBOT	ViT-B/16	250	69.7
DINO	ViT-B/8	300	70.0
SimCLR v2	RN151 (2x)	800	70.2
BYOL	RN200 (2x)	800	71.2
MSN	ViT-B/4	300	75.7

Low Shot Classification Accuracy: Semi-supervised evaluation on ImageNet-1k with 1% of the labels (approximately 12 labeled images per class). Source: [Meta I-JEPA blog](#), 2023; data from Table 2 of Assran et al., CVPR 2023.

- ▶ Strong **off-the-shelf** representations: linear probe, semi-supervised, and competitive low-shot settings without extensive task-specific fine-tuning of the backbone.
- ▶ Transfers to classification, detection, segmentation-style use cases; blog and paper emphasize **broader task mix** including some low-level vision tasks.
- ▶ Open direction: extend JEPAs to **richer modalities** (video, audio, text conditioning) for longer-horizon **world models**.

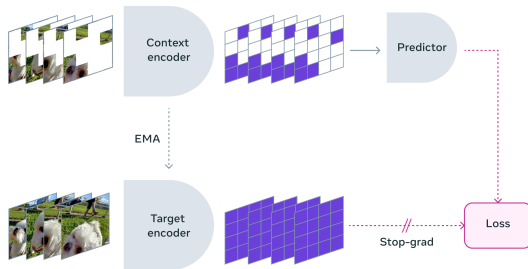
- ▶ The visual world is dynamic: understanding actions and events requires temporal structure.
- ▶ **V-JEPA** (Video Joint Embedding Predictive Architecture, Meta) extends the JEPA idea to video: predict *representations* of masked spatio-temporal content, not pixels.
- ▶ Like I-JEPA, it is **non-generative**: learning targets live in a learned latent space, so the model can ignore unpredictable detail.
- ▶ Pre-training uses **unlabeled video**; labeled data is reserved for lightweight adaptation.



Decorative Meta AI illustration (hero art of the [MuAViC blog post](#), 2023); not a V-JEPA figure.

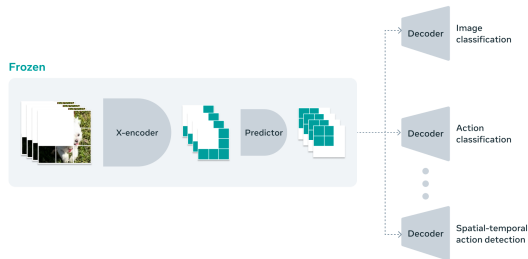
- ▶ Masking must be hard enough: **sparse random patches** make the task too easy, and little is learned about dynamics.
- ▶ If a region is masked only at **one instant** while adjacent frames stay visible, the model can “cheat” from neighboring time steps.
- ▶ V-JEPA uses masking across **both space and time** so the model must infer what happens in the scene rather than copy texture from nearby frames.

- ▶ **Context path:** encodes visible video patches; **predictor** maps to the target representation space (JEPA-style).
- ▶ **Target path:** an EMA encoder processes the full unmasked clip; targets are its output tokens at the masked positions (stop-gradient). Loss matches predicted and target embeddings.
- ▶ Same high-level recipe as I-JEPA, extended to **3D** patch tokens over space and time.



Reference: [Meta V-JEPA blog](#), 2024 (blog graphic).

- ▶ After self-supervised pre-training, the **encoder and predictor are frozen**.
- ▶ Same frozen backbone can support several tasks (e.g., action classification, fine-grained interaction recognition, activity localization) without full fine-tuning that would specialize the whole network to one benchmark.



Reusing frozen representations for downstream video understanding.

Reference: [Meta V-JEPA blog](#), 2024 (blog graphic).

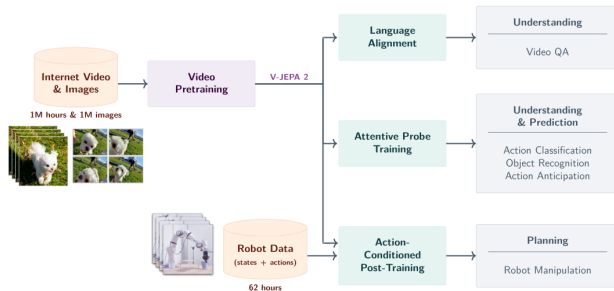
- ▶ Strong on **frozen** benchmarks spanning image/video classification, action recognition, and spatio-temporal action detection relative to prior video representation learning.
- ▶ Particularly strong on **fine-grained** interaction understanding over short clips (e.g., distinguishing put-down vs. pick-up vs. pretend actions).
- ▶ Reported limitations include **short clip** focus (on the order of seconds); **audio** and **long-horizon** prediction/planning highlighted as future directions.

Method	Arch.	Frozen Evaluation					
		K400 (16×8×3)			SSv2 (16×2×3)		
		5%	10%	50%	5%	10%	50%
MVD	ViT-L/16	62.6 ± 0.2	68.3 ± 0.2	77.2 ± 0.3	42.9 ± 0.8	49.5 ± 0.6	61.0 ± 0.2
VideoMAE	ViT-H/16	62.3 ± 0.3	68.5 ± 0.2	78.2 ± 0.1	41.4 ± 0.8	48.1 ± 0.2	60.5 ± 0.4
VideoMAEv2	ViT-g/14	37.0 ± 0.3	48.8 ± 0.4	67.8 ± 0.1	28.0 ± 1.0	37.3 ± 0.3	54.0 ± 0.3
V-JEPA	ViT-H/16 ₃₈₄	68.2 ± 0.2	72.8 ± 0.2	80.6 ± 0.2	54.0 ± 0.2	59.3 ± 0.5	67.9 ± 0.2

Low-shot frozen evaluation on Kinetics-400 and Something-Something-v2 (Meta V-JEPA blog; [Bardes et al.](#)).

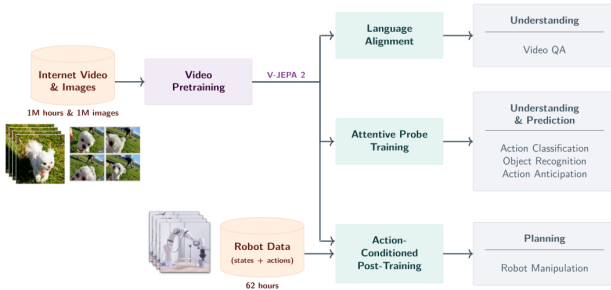
Same **visual mask denoising** / JEPA spirit as I-JEPA, but at **internet scale**.

- ▶ Order of **~1M hours** of video and **~1M images** for pre-training.
- ▶ Downstream use: video understanding, video QA, and planning.



Assran et al. (2025): overview of pre-training, alignment, and robotics post-training.

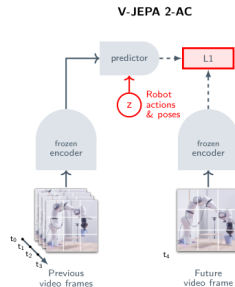
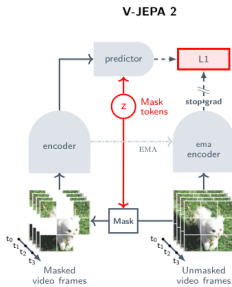
- ▶ Stage **A**: SSL on web video/images → strong motion and appearance understanding.
- ▶ Stage **B**: **language alignment**: project video tokens into an LLM through a small MLP projector so the backbone supports VQA and reasoning.
- ▶ Stage **C**: optional **action-conditioned** head (V-JEPA 2-AC) on tens of hours of robot data for latent planning.



Assran et al. (2025), Fig. 1: overview of pre-training, alignment, and robotics post-training.

- ▶ **Visual Question Answering (VQA):** answer natural-language questions about an image or video (“What is the person about to pick up?”).
- ▶ Standard recipe (LLaVA-style): keep the **vision encoder frozen** (or lightly tuned), train a small **projector** that maps visual tokens into the LLM’s embedding space, then instruction-tune on paired visual–text data.
- ▶ V-JEPA 2 follows this pattern: a small **MLP projector** maps the encoder’s video tokens into the LLM’s embedding space ($\sim 8\text{B}$ -parameter Llama backbone), and for the best results the coupled stack, *including the encoder*, is then trained end-to-end on video–text data.
- ▶ Note that the JEPA encoder is **pre-trained without any language supervision**; grounding is added *afterwards* (projector warm-up, then instruction tuning that also updates the encoder).

V-JEPA 2: Training Stage 1 (Pre-training)

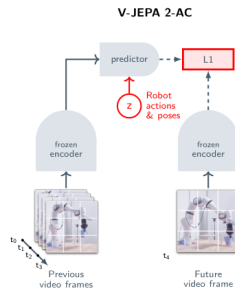
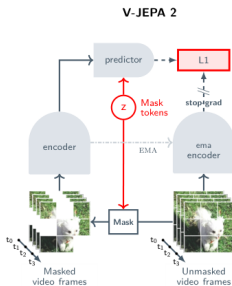


Assran et al. (2025): multistage training.

► V-JEPA 2:

- Video clips divided into tokens; a subset is masked.
- Encoder processes masked sequence to produce token embeddings.
- Mask tokens concatenated with encoder outputs.
- Predictor regresses to targets computed by an EMA encoder using L_1 loss.

V-JEPA 2: Training Stage 2 (Action-Conditioned)



Assran et al. (2025): multistage training.

► V-JEPA 2-AC:

- Freeze pretrained encoder and train action-conditioned predictor on top (<62 hours of unlabeled robot trajectories).
- Use autoregressive prediction to model future frame representations based on past frames, actions, and end-effector states.

Once the action-conditioned predictor is trained, it works as a *simulator in representation space*, and **planning** requires no task-specific training.

1. **Specify the goal as an image**; encode it to a goal representation s_g .
 2. **Sample candidate action sequences**; roll each out through the predictor to get predicted future representations.
 3. **Score** each rollout by the (L_1) distance between its predicted final state and s_g ; refine the action distribution toward the best rollouts (**cross-entropy method, CEM**).
 4. **Execute only the first action**, observe, and **re-plan**: receding-horizon **model predictive control**, i.e. the multistep world-model loop shown earlier.
- Result: **zero-shot** pick-and-place with Franka arms in labs **never seen in training**, with no rewards and no demonstrations in the target environment (as claimed in the paper).

TASK TYPE	BENCHMARK	V-JEPA 2	PREVIOUS BEST
Planning and Robot Control from Image Goals	Reach	100%	100% (Octo)
	Grasp	45%	8% (Octo)
	Pick-and-place	73%	13% (Octo)
Prediction	Epic-Kitchens-100 action anticipation	39.7%	27.6% (PlausiVL)
Understanding	Something-Somethingv2 action recognition Attentive probe	77.3%	69.7% (InternVideo2-1B Attentive probe)
	Diving48 Attentive probe	90.2%	86.4% (InternVideo2-1B Attentive probe)
	Perception Test	84.0%	82.7% (PerceptionLM)
	MVPBench	44.5%	39.9% (InternVL-2.5)

Meta AI, [V-JEPA research page](#) (2025): V-JEPA 2 performance summary; underlying numbers from Assran et al. (2025).

- ▶ **Fine-grained motion:** Something-Something v2 top-1 **77.3** (paper-reported).
- ▶ **Action anticipation:** Epic-Kitchens-100 recall@5 **39.7**, described as strong vs. prior task-specialized models.
- ▶ **Video QA (after LLM alignment):** e.g. PerceptionTest **84.0**, TempCompass **76.9** at the **8B-parameter scale** (ViT-g encoder + Llama 3.1 8B LLM; paper-reported).
- ▶ **Robotics:** V-JEPA 2-AC trained on **<62 hours** of Droid-style trajectories; **zero-shot** Franka pick-place demos in **unseen labs** without environment-specific teleoperation or task rewards (as claimed in the paper abstract).

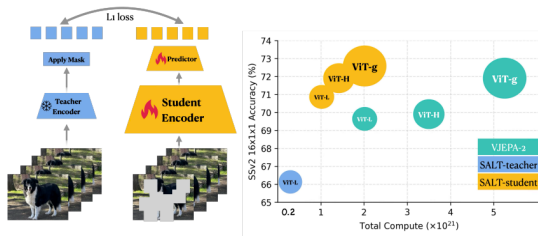
Treat numbers as **snapshot** results from Assran et al.; see [PDF](#) for protocols and ablations.

- ▶ **Point-JEPA:** applies the JEPA recipe to **unordered 3D point sets**; uses spatial proximity and masking in the point domain so the model predicts **latent structure** of masked regions from geometric context (not pixels).
- ▶ **CNN-JEPA:** keeps **convolutional** encoders (sparse, translation-friendly) while retaining **latent prediction** rather than pixel reconstruction; useful when ViTs are not the deployment backbone.
- ▶ Related 3D SSL: **Sonata** (2025) pursues highly reliable point representations from large-scale self-supervision; complementary reading for geometry-heavy applications.
- ▶ **VL-JEPA** extends the same **predictive latent** philosophy to **vision–language** setups: build world models that couple visual structure with language for richer situational understanding.

- ▶ **MC-JEPA** (Bardes et al., 2023): learns **motion** and **content** jointly with a *shared* encoder. One branch estimates **optical flow** between frames (a JEPA-style prediction task in feature space), the other learns content features (VICReg-style regularization); each objective improves the other.
- ▶ **A-JEPA** (Fei et al., 2023): applies the masked latent-prediction recipe to **audio**: mel-spectrograms are patchified like images, and the model predicts representations of masked time–frequency regions (with a curriculum masking strategy).
- ▶ The JEPA recipe is **modality-agnostic**: masking plus latent prediction applies to images, video, point clouds, audio, and motion.

- ▶ **SALT** (*Static-teacher Asymmetric Latent Training*, Li et al., 2025) uses a **two-stage recipe**: (i) train a teacher with a **pixel reconstruction** objective under V-JEPA-style masking; (ii) **freeze** the teacher and train a student to predict teacher latents on masked regions.
- ▶ Reported benefits: simpler optimization than coupled EMA schedules, favorable **accuracy–compute** trade-offs in frozen evaluation, and student quality surprisingly robust to teacher size.

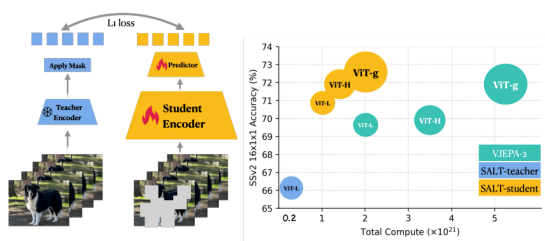
SALT: Two-Stage Recipe for Video SSL (Stage 1)



Reference: Li et al. (2025) SALT, [arXiv:2509.24317](https://arxiv.org/abs/2509.24317), Fig. 1.

- **Goal:** Train a **teacher encoder** to robustly reconstruct masked video regions *at the pixel level*.
- **Approach:** Use a **V-JEPA-style masking** scheme; teach the encoder to infill missing pixels based on visible context.
- **Outcome:** Teacher encoder learns to summarize temporal and spatial structure, producing feature-rich latent representations.

SALT: Two-Stage Recipe for Video SSL (Stage 2)



Reference: Li et al. (2025) SALT, [arXiv:2509.24317](https://arxiv.org/abs/2509.24317), Fig. 1.

- **Freeze** the teacher encoder weights.
- Train a student to **predict teacher latent embeddings** of masked regions, using only visible context as input.
- Mimics standard JEPA: loss is in **latent space**, not directly on pixels.

- ▶ **Full fine-tuning gap:** JEPA's advantage shows in **frozen** and **low-shot** probes; with full fine-tuning, reconstruction methods like MAE remain highly competitive on many benchmarks.
- ▶ **Masking is a hand-tuned inductive bias:** block sizes, ratios, and spatio-temporal shapes must be re-designed per modality; the recipe still involves substantial engineering.
- ▶ **EMA fragility:** coupled teacher–student schedules add hyperparameters and instability; this is the issue SALT's frozen-teacher recipe targets.
- ▶ **Reconstruction may not be wasted:** generative models such as diffusion also learn strong representations; the claim that pixel-level losses waste capacity is a hypothesis rather than a settled result.
- ▶ **V-JEPA 2's own reported limitations:** sensitivity to camera pose, drift over long planning horizons, and goals specified only as **images** (no language-conditioned planning).
- ▶ **Source bias:** much of the JEPA narrative comes from a single lab (Meta / LeCun); read the claims and the emphasized benchmarks critically.

- ▶ **JEPA: A new paradigm for Self-Supervised Learning**
Joint Embedding Predictive Architectures (JEPA) learn *internal world models* by predicting **abstract representations** of masked or future regions, instead of raw data, enabling robust and efficient learning.
- ▶ **Contrast to Classic SSL:** JEPA unifies ideas from contrastive (invariant) and generative SSL, but avoids pixel-level prediction.
- ▶ **Strong Transfer:** Predicting stable, high-level embedding targets yields representations that excel at downstream tasks such as VQA, reasoning, and planning.
- ▶ **Flexible Recipe:**
The JEPA framework extends to images (I-JEPA), video (V-JEPA, V-JEPA 2), point clouds (Point-JEPA), convolutional backbones (CNN-JEPA), and multimodal data (VL-JEPA).
- ▶ **Active Areas:** Ongoing work explores efficient teachers (SALT), hierarchical and recurrent world models, multimodal fusion, and theoretical understanding of latent prediction.

- ▶ **Masking and targets:** How much to mask, where, and at what semantic scale (patch vs. object vs. clip)?
- ▶ **Long horizon:** Original V-JEPA focused on short clips; **V-JEPA 2** pushes scale and adds action-conditioned planning (V-JEPA 2-AC), but long-horizon world models remain open.
- ▶ **Multimodal fusion:** aligning audio, language, and video under JEPA-style latent prediction without collapsing modalities.
- ▶ **Theory:** when does latent prediction provably recover predictable structure, and when do shortcuts dominate?
- ▶ **Deployment:** latency, memory, and hardware-aware training for billion-scale video corpora.

- ▶ **Stanford CS231n**, Lecture 12 (SSL history, pretext tasks, linear probing): [slides](#).
- ▶ **Berkeley CS294-158**, Deep Unsupervised Learning (likelihood models → non-generative SSL): [SP20 site](#).
- ▶ **CMU 11-785** Intro to Deep Learning / **DLSys** lecture hub: [deeplearning.cs.cmu.edu](#), [dlsyscourse.org](#).
- ▶ **MIT 6.S191** Introduction to Deep Learning: [introtodeeplearning.com](#).
- ▶ **Survey (ViT + SSL)**: mechanisms taxonomy for vision transformers, [arXiv:2408.17059](#).
- ▶ **LeCun lecture deck** (Les Houches / world-model narrative): [PDF](#).

1. Assran et al., “Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture,” CVPR 2023. [arXiv:2301.08243](#)
2. Bardes et al., “Revisiting Feature Prediction for Learning Visual Representations from Video,” 2024. [arXiv:2404.08471](#)
3. LeCun, “A Path Towards Autonomous Machine Intelligence,” OpenReview 2022. [OpenReview](#)
4. Meta AI, “I-JEPA: The first AI model based on Yann LeCun’s vision for more human-like AI,” *AI at Meta* blog, 2023. [ai.meta.com](#)
5. Meta AI, “Yann LeCun on a vision to make AI systems learn and reason like animals and humans,” *AI at Meta* blog. [ai.meta.com](#)
6. Meta AI, “Introducing V-JEPA 2,” V-JEPA research hub. [ai.meta.com/research/vjepa](#)
7. Assran et al., “V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning,” 2025. [arXiv:2506.09985](#)
8. “Tutorial on Joint Embedding Predictive Architectures (JEPA),” TechRxiv. [DOI:10.36227/techrxiv.176469421.19270944](#)
9. Kalapos & Gyires-Tóth, “CNN-JEPA: Self-Supervised Pretraining Convolutional Neural Networks Using Joint Embedding Predictive Architecture,” 2024. [arXiv:2408.07514](#)
10. Saito et al., “Point-JEPA: A Joint Embedding Predictive Architecture for Self-Supervised Learning on Point Cloud,” 2024. [arXiv:2404.16432](#)

11. Wu et al., “Sonata: Self-Supervised Learning of Reliable Point Representations,” CVPR 2025.
[arXiv:2503.16429](#)
12. Li et al., “Rethinking JEPA: Compute-Efficient Video SSL with Frozen Teachers” (SALT), ICLR 2026 (Apple ML Research). [arXiv:2509.24317](#)
13. Khan et al., “A Survey of the Self Supervised Learning Mechanisms for Vision Transformers,” 2024.
[arXiv:2408.17059](#)
14. Meta AI, “V-JEPA: The next step toward advanced machine intelligence,” *AI at Meta* blog, 2024.
[ai.meta.com](#)
15. Bardes, Ponce & LeCun, “MC-JEPA: A Joint-Embedding Predictive Architecture for Self-Supervised Learning of Motion and Content Features,” 2023. [arXiv:2307.12698](#)
16. Fei et al., “A-JEPA: Joint-Embedding Predictive Architecture Can Listen,” 2023. [arXiv:2311.15830](#)
17. Chen et al., “VL-JEPA: Joint Embedding Predictive Architecture for Vision-language,” 2025.
[arXiv:2512.10942](#)
18. He et al., “Masked Autoencoders Are Scalable Vision Learners,” CVPR 2022.
19. Bao et al., “BEiT: BERT Pre-Training of Image Transformers,” ICLR 2022.
20. Baevski et al., “data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language,” ICML 2022.
21. Chen et al., “A Simple Framework for Contrastive Learning of Visual Representations (SimCLR),” ICML 2020.

22. Grill et al., “Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning (BYOL),” NeurIPS 2020.
23. Caron et al., “Emerging Properties in Self-Supervised Vision Transformers (DINO),” ICCV 2021.
24. Zhou et al., “iBOT: Image BERT Pre-Training with Online Tokenizer,” ICLR 2022.
25. Bardes, Ponce & LeCun, “VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning,” ICLR 2022.
26. Garrido et al., “On the Duality Between Contrastive and Non-Contrastive Self-Supervised Learning,” ICLR 2023. [arXiv:2206.02574](https://arxiv.org/abs/2206.02574)
27. Devlin et al., “BERT,” NAACL 2019.
28. Dosovitskiy et al., “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale (ViT),” ICLR 2021.
29. Stanford CS231n, Lecture 12 slides (SSL). [PDF](#)
30. UC Berkeley CS294-158 (SP20), Deep Unsupervised Learning. [course site](#)
31. CMU 11-785 Deep Learning. deeplearning.cs.cmu.edu
32. MIT 6.S191 Introduction to Deep Learning. introtodeeplearning.com
33. Vert, A. & Kuka, V., “What Is JEPA? LeCun Architecture & World Models,” *Turing Post*, 2026. turingpost.com
34. LeCun, Les Houches lecture slides (AMI / world models). [PDF](#)

- 35. LeCun, "From Machine Learning to Autonomous Intelligence / Objective-Driven AI," LMU München & Bayerische Akademie der Wissenschaften, 29 Sep 2023 (lecture slides).
- 36. LeCun, "Objective-Driven AI: Towards AI systems that can learn, remember, reason, plan, have common sense, yet are steerable and safe," Ding-Shum Lecture, Harvard University, 28 Mar 2024 (lecture slides).